

ARQILA SURYA PUTRA

📍 Bandung, Indonesia ✉ arqilasp@gmail.com 🌐 arqilasp.com
🌐 linkedin.com/in/arqila-surya-putra 📄 github.com/pablonification

Full Stack Engineer specializing in Payment Systems, Scalable Microservices, and High-Traffic Web Applications

EDUCATION

Institut Teknologi Bandung (ITB)

Bandung, Indonesia

Bachelor of Engineering in Information and System Technology

2023 – 2027

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Database Systems, Software Engineering.
- **Awards:** 1st Runner Up AI Innovation Challenge (COMPFEST 17), Best Innovation Award (ILAS 2024).

EXPERIENCE

ITBFess

Sep 2024 – Present

Founding Engineer (Full Stack)

Bandung, West Java

- Integrated **Xendit payment gateway** for premium feature monetization, implementing webhook handlers, idempotent transaction processing, and payment state management in production.
- Scaled a viral social platform (24,000+ X followers, **1M+ weekly impressions**) to handle **1,800+ MAU** and **90,000+ monthly events** with zero downtime.
- Built Node.js backend with Express handling real-time Socket.IO connections, async job queues for Twitter API integration, and MongoDB persistence.
- Implemented rate limiting, abuse prevention, and token-based economy system processing real financial transactions.

Containder Indonesia

Apr 2025 – Aug 2025

AI Engineer Intern

Remote

- Developed a smart recycling prototype using **YOLOv5** and **OpenCV** to classify and estimate plastic bottle volume on edge devices.
- Optimized computer vision architectures for resource-constrained edge environments by fine-tuning model parameters and input processing, successfully achieving real-time inference speeds required for immediate physical feedback.
- Conducted comparative analysis of detection algorithms (ArUco vs. Object Detection) to determine optimal production architecture.

8EH Radio ITB

Apr 2025 – Present

Technology & Media Manager

Bandung, West Java

- Engineered a **RAG-based AI chatbot** and automated article summarization pipeline, modernizing the user content consumption experience.
- Overhauled legacy WordPress templates into a bespoke **Next.js** interface, delivering a high-fidelity, intuitive UI/UX that aligns with modern design standards.
- Built a custom **Shoutcast ICY stream proxy** within Next.js API routes to resolve mixed-content (HTTP/HTTPS) issues and handle CORS for legacy radio protocols.
- Implemented a hybrid storage strategy using **Cloudflare R2** for high-availability podcast hosting, reducing bandwidth costs while ensuring low-latency delivery.

PROJECTS

Tukangin | Next.js, TypeScript, Prisma, PostgreSQL, Supabase

Nov 2024 – Present

- Designed **microservices architecture** with 4 core services (Ordering, Dispatch, Execution, Post-Service) following SOA/ML contracts.
- Achieved **100% test coverage** (29/29 assertions) with Jest integration testing across all service boundaries.

Meddocs Community Platform | Next.js, Node.js, MongoDB, Midtrans, Twilio

Jun 2025 – Aug 2025

- Architected a secure membership portal for 100+ medical professionals, implementing strict **Role-Based Access Control (RBAC)**.
- Engineered end-to-end monetization and notification pipelines by integrating **Midtrans** payment gateways and **Twilio** SMS services.
- Optimized media storage and delivery for mobile clients using **Cloudinary**, ensuring low-latency access to medical resources.

Setorin - AI Recycling System | Next.js, FastAPI, MongoDB, YOLO, WebSocket

Apr 2025

- **1st Runner Up** at AI Innovation Challenge COMPFEST 17 – Built AI-powered recycling system with real-time bottle detection.
- Implemented reward system with **e-wallet withdrawal integration** and automated point calculation based on volume/brand detection.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Java, C, SQL

Backend & APIs: Node.js, Express, REST APIs, WebSocket, Webhook Handling, Job Queues

Databases & Cloud: MongoDB, PostgreSQL, Prisma, Firebase, Supabase, Vercel, DigitalOcean, AWS

Frontend: Next.js, React, React Native, Expo, Tailwind CSS

Payment & Integrations: Xendit, Midtrans, TripayID, Twilio, Resend

DevOps & Tools: Git, Docker, PM2, Nginx, CI/CD, Jest

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate (Sep 2025)

EF SET English Certificate – C2 Proficient (71/100) (Mar 2025)